



*COMP7503 Multimedia Technologies*

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# Introduction

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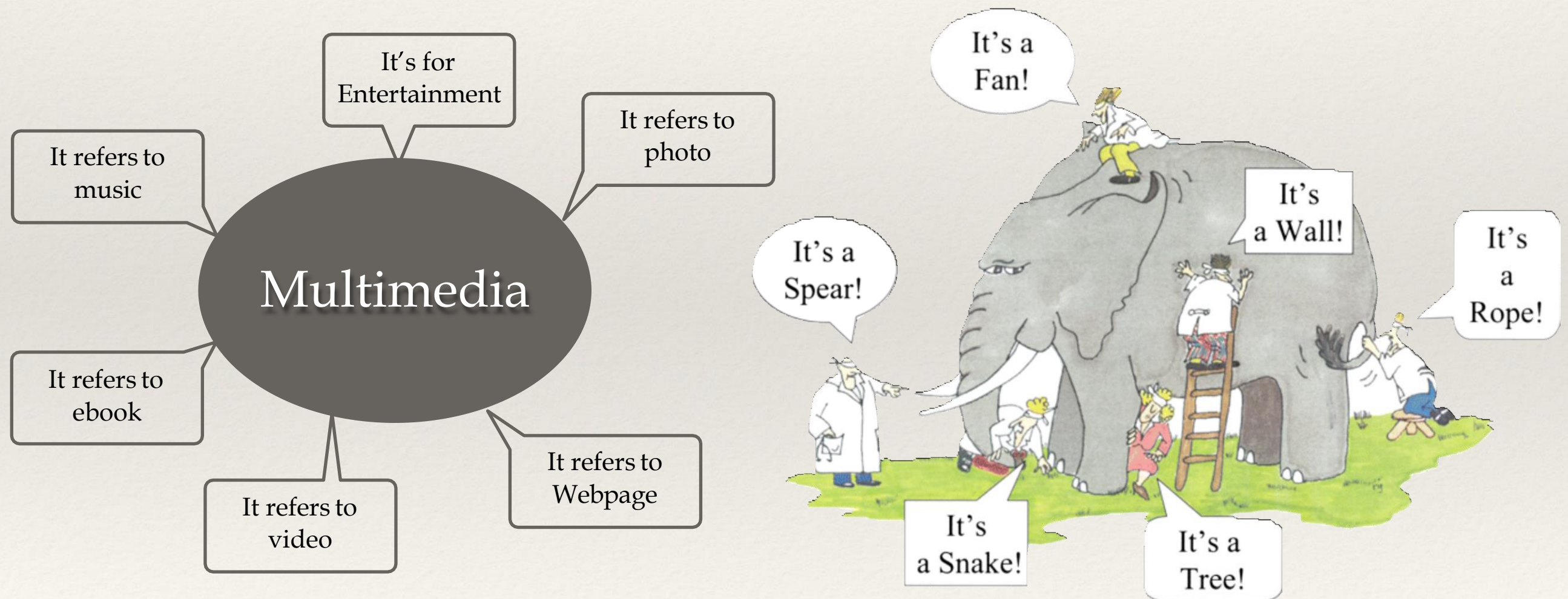
THE UNIVERSITY OF HONG KONG

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D E P A R T M E N T O F  
**COMPUTER SCIENCE**

# What is Multimedia?

- ❖ Everybody thinks they know about the word “multimedia”, but most of them think of multimedia from narrow perspectives



# Is audio alone enough?



Video adopted from <https://www.youtube.com/watch?v=FeTOf49mUyA>



# Is text information alone enough?

- ❖ Under which scenario a person would say the following
  - ❖ “Just lie down, shut up, and open wide”



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# Is text information alone enough?

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- ❖ Under which scenario a person would say the following
  - ❖ “Just lie down, shut up, and open wide”



It's because he/she is a dentist



# Why Multimedia?

- ❖ How do humans receive information?

- ❖ Five senses

- ❖ Sight
    - ❖ Hearing
    - ❖ Taste
    - ❖ Smell
    - ❖ Touch



- ❖ Each sense provides only partial information
- ❖ Multimedia refers to the combined use of several media that stimulate one or more of these five senses to deliver relevant information to users



# Why Multimedia? (Cont'd)

- ❖ Audio and Visual Information dominates human activities
  - ❖ Humans are equipped with powerful visual and audio analysis capability
  - ❖ Taste, smell, and touch cannot be easily acquired and reproduced by existing multimedia computing technologies



# Then, What is computing?

- ❖ Computing is any goal-oriented activity requiring, benefiting from or creating computing machinery.

(<https://en.wikipedia.org/wiki/Computing>)

- ❖ Multimedia computing means utilise signals from multifarious sensors and present to users only the relevant information in the appropriate sensory modality (media)

- ❖ The goal of multimedia computing
  - ❖ Information sharing on experiences
  - ❖ Knowledge extracting from experiences



# Communications in Human Society

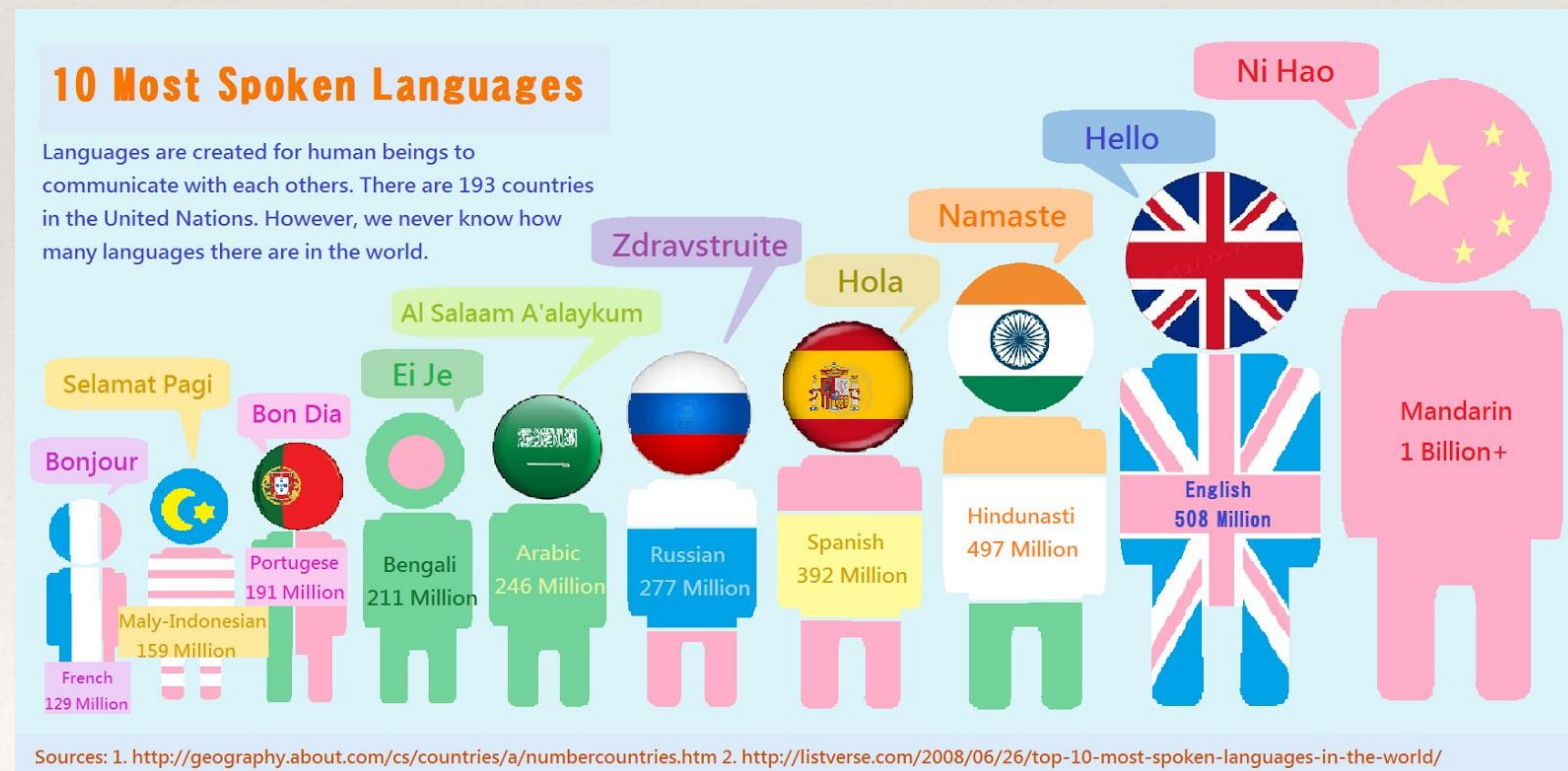
- ❖ The Communication is key for us to
  - ❖ Share experiences
  - ❖ Create, maintain, sustain, and propagate knowledge
- ❖ Communication across **space**
  - ❖ Allows people to exchange information independent of their current locations
  - ❖ Examples? What are the technologies required to support this?
- ❖ Communication across **time**
  - ❖ Allows people to experience an event over and over again without having to be there at the exact moment
  - ❖ Examples? What are the technologies required to support this?
- ❖ Communication to as many people as possible (**addressee**)
  - ❖ Examples? What are the technologies required to support this?



# Human Communication Means (1)

## ❖ Spoken Languages

- ❖ Consisted of analog sounds uttered with the speech-producing infrastructure in the throat
- ❖ Able to convey complex facts
- ❖ Any invariance across time, space, and addressee?



# Human Communication Means (2)

## ❖ Written Languages

- ❖ Evolved from drawings and paintings
- ❖ Enables sharing of experiences and knowledge
- ❖ Initial bulky preservation techniques (e.g. stone tablets) evolved into more practical storage devices and writing methods
  - ❖ e.g. Paper and ink
- ❖ Any invariance across time, space, and addressee?

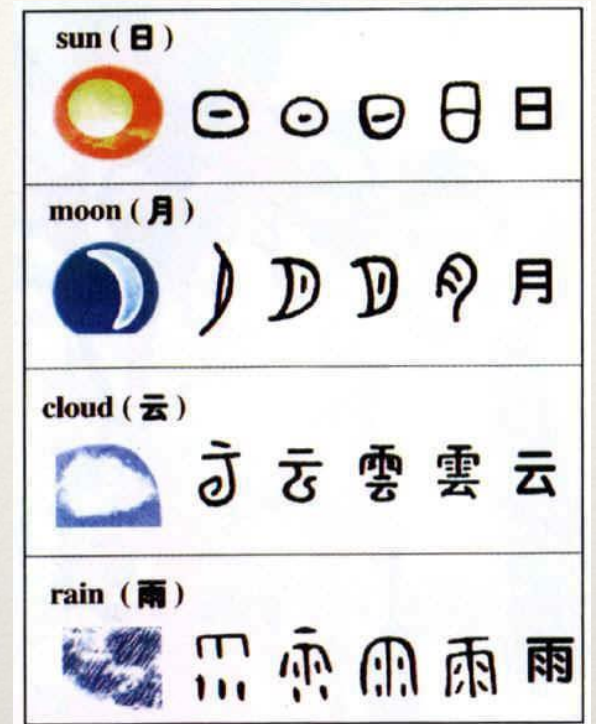


Image adopted from [http://blog.sina.com.cn/s/blog\\_5d8bca8a0101e0ef.html](http://blog.sina.com.cn/s/blog_5d8bca8a0101e0ef.html)

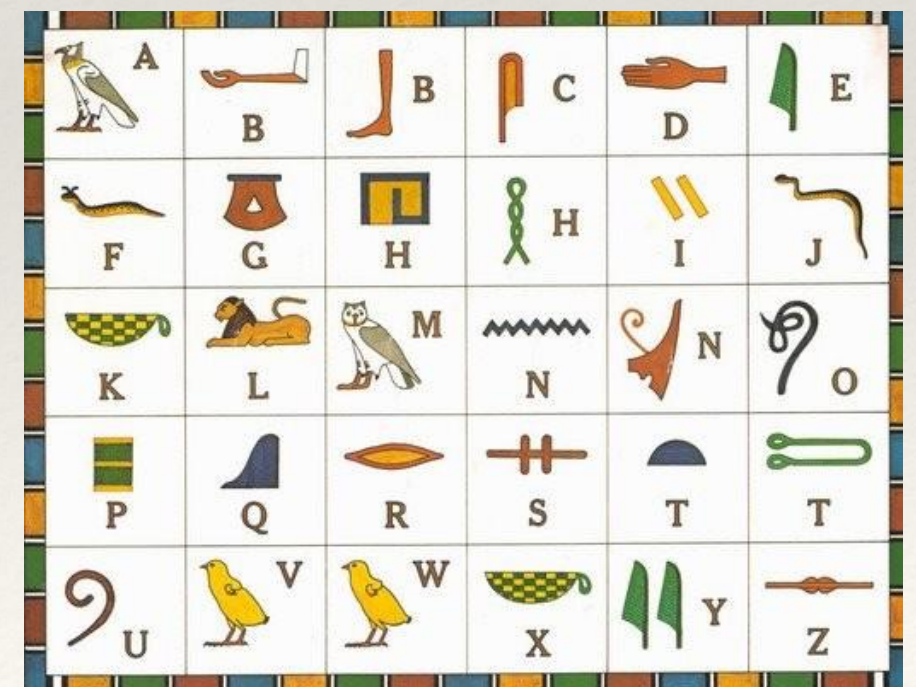


Image adopted from <http://ahmed-maher.blogspot.hk>



# Human Communication Means (3)

## ❖ Printing Press

- ❖ Invented by Johannes Gutenberg around year 1440
- ❖ Enabled mass communication by making creation, storage, and distribution of documents more manageable
- ❖ Makes information and knowledge sharing earlier
- ❖ Any invariance across time, space, and addressee?

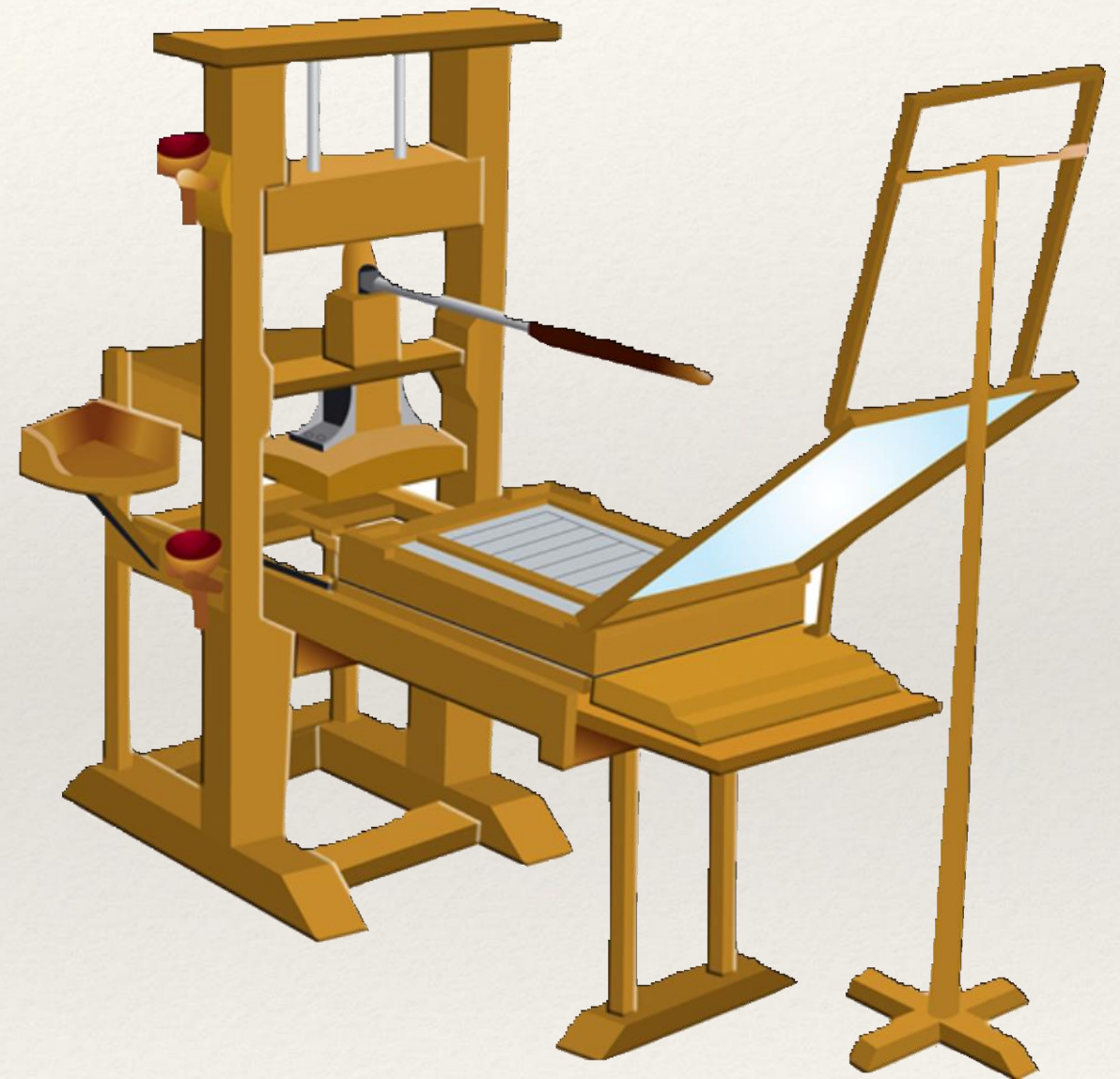


Image acquired from <http://www.howitworksdaily.com/see-inside-the-gutenberg-printing-press/>



# Human Communication Means (4)

## ❖ Telegraph

- ❖ Allows instantaneous communication of symbolic information over long distances
- ❖ Any invariance across time, space, and addressee?

## ❖ Telephones

- ❖ Allows natural communication (talking) over long distance
- ❖ Any invariance across time, space, and addressee?

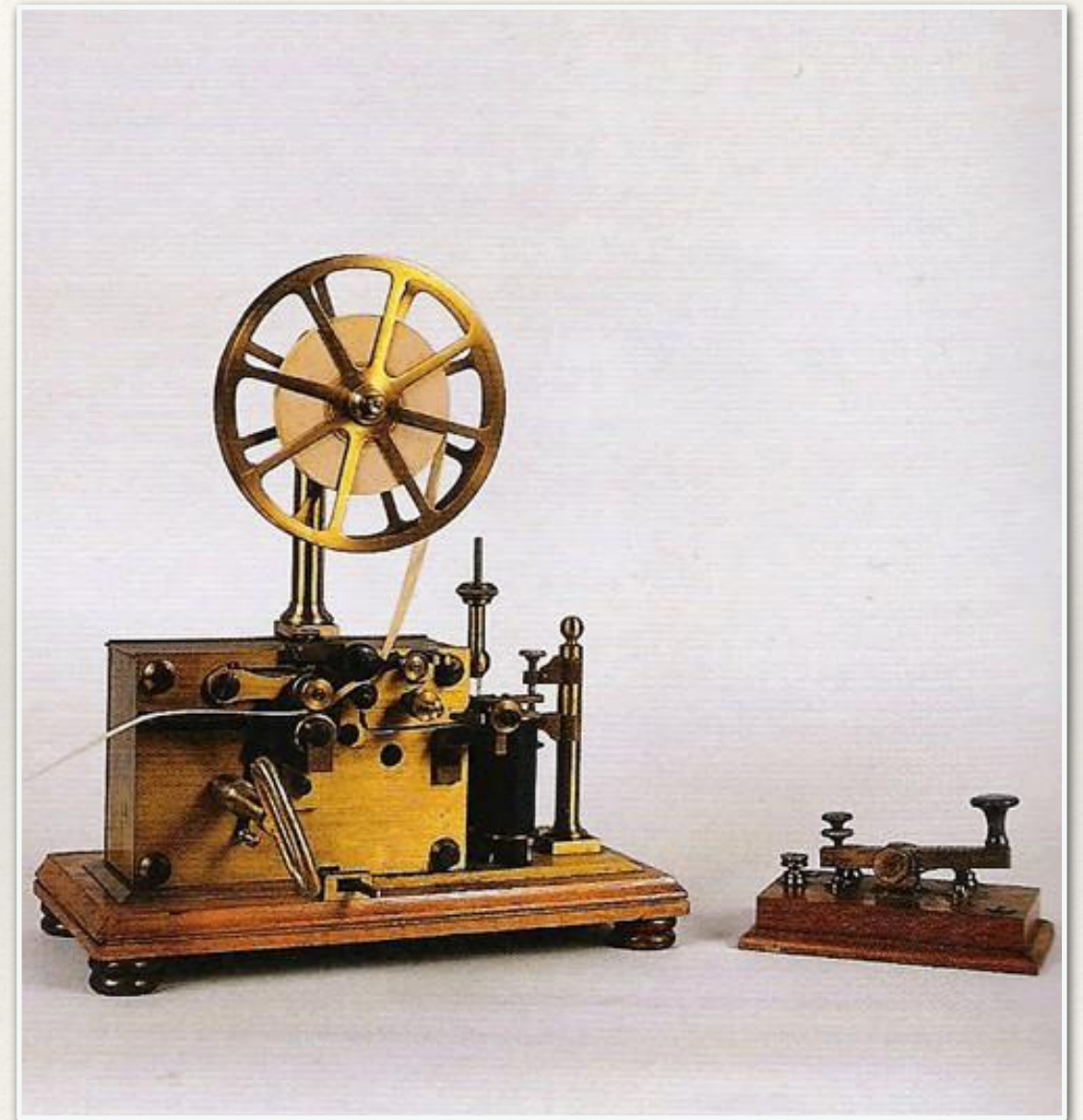


Image acquired from <http://imgbuddy.com>



# Human Communication Means (5)

## ❖ Radio

- ❖ Allows distribution of sound wirelessly
- ❖ First form of instantaneous mass communication
- ❖ Any invariance across time, space, and addressee?



Image acquired from <http://radiomunansi.info/about-us/>

## ❖ Movies and Television

- ❖ Combine our senses of sight and hearing for more natural communication
- ❖ Effective way to key into our emotions
- ❖ Any invariance across time, space, and addressee?



Image acquired from [https://fbcpolson.files.wordpress.com/2014/08/television\\_1.jpg](https://fbcpolson.files.wordpress.com/2014/08/television_1.jpg)



# Human Communication Means (6)

## ❖ Storage Technologies

### ❖ Storage Media

- ❖ Magnetic Tape
- ❖ Digital Storage

❖ Allows storage, preservation and distribution of sound, video, etc.

❖ Any invariance across time, space, and addressee?

## ❖ Distribution Technologies

❖ Sending physical copy

❖ Copy by wire

❖ Internet distribution

❖ Any invariance across time, space, and addressee?



Images acquired from  
<http://www.bbc.co.uk/schools/gcsebitesize/ict/hardware/1datastoragerev6.shtml>



Image acquired from  
[http://www.dealersolutions.com.au/car\\_inventory\\_distribution?content=overview](http://www.dealersolutions.com.au/car_inventory_distribution?content=overview)



# Communications-related Inventions

| Inventions        | Resulting Applicatoin              | Invariance               |
|-------------------|------------------------------------|--------------------------|
| Spoken Languages  | Natural communication              | None                     |
| Written Languages | Symbolic record of language        | Time                     |
| Paper             | Portability                        | Time, space              |
| Print             | Mass distribution                  | Time, space, addressee   |
| Telegraph         | Remote narrow communication        | Space                    |
| Telephone         | Remote analog communication        | Space                    |
| Radio             | Broadcasting of sound              | Space, addressee         |
| Television        | Broadcasting of sight and sound    | Space, addressee         |
| Movies            | Recording of sight and sound       | Time, space, addressee   |
| Recording media   | Recording                          | Time, space, addressee   |
| Digital media     | Machine enhancement and processing | Time, space, addressee   |
| Internet          | Personalized reception             | Time, space, (addressee) |



# The role of Computer in Multimedia during 1960s-era

- ❖ During 1960s, computer is bulky
  - ❖ e.g. IBM1620
  - ❖ No ALU, arithmetics processing is done based on lookup tables
  - ❖ Hardware
    - ❖ CPU a.k.a. *CADET*
      - ❖ *Can't Add, Doesn't Even Try*
    - ❖ RAM: 60kbytes
    - ❖ I/O: Paper Tape reader and writer, or Punch Cards
  - ❖ Software
    - ❖ Symbolic Programming System (assembly language)
    - ❖ Fortran



**A 1960s-era Computer at NASA**

Image adopted from  
Figure 2.1 in the textbook



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# The role of Computer in Multimedia

## Before 1990s

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- ❖ Before 1990s, computing power, storage, bandwidth, and processing algorithms were not capable to deal with audio and video
  - ❖ Using multimedia in computer was under R&D phase
  - ❖ Hardware
    - ❖ CPU < Intel 80286, < 6MHz
    - ❖ RAM < 640KB
    - ❖ Graphics Modes: 320x200 / 640x200, 16 colors
    - ❖ Storage: Floppy Drive, Harddisk <= 20MB
  - ❖ Software
    - ❖ OS: MS-DOS
    - ❖ Mostly text-based applications



PC XT 286

Image adopted from

<http://www.old-computers.com/museum/computer.asp?st=1&c=260>



# The role of Computer in Multimedia

## Early 1990s

- ❖ Early 1990s, high end PC's are equipped with multimedia capabilities, like IBM PS/2
  - ❖ Hardware
    - ❖ CPU Intel 80386/80486, 50-60 MHz
    - ❖ RAM  $\leq 64\text{MB}$
    - ❖ Graphics Modes:  $640 \times 480$ ,  $1024 \times 768$ , 256/64k colors
    - ❖ Storage: Floppy Drive, Harddisk  $\leq 400\text{MB}$
    - ❖ May be equipped with dedicated Sound Card (e.g. Soundblaster)
  - ❖ Software
    - ❖ OS: MS-DOS
    - ❖ GUI crafted by Application
    - ❖ 3D Gaming Possible



IBM PS/2



3D Games Wolfenstein on  
386/486 PC



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# The role of Computer in Multimedia

## Mid 1990s

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- ❖ Around mid 1990s, more high end PC's evolved, like Acer Presario 4122
  - ❖ Hardware
    - ❖ CPU Intel Pentium 150MHz
    - ❖ RAM up to 128MB
    - ❖ Graphics Modes: up to 1280x1024
    - ❖ Storage: Floppy Drive, 2.5GB harddrive, 8x CD-ROM
    - ❖ Integrated 16-bit Stereo Sound
  - ❖ Software
    - ❖ OS: Windows 95
    - ❖ Audio Player, Video Player
    - ❖ CorelDraw 5
    - ❖ Various kinds of games



Compaq Presario 4122

Image adopted from <http://www.computinghistory.org.uk/det/5043/Compaq-Presario-4122/>



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## The role of Computer in Multimedia by the end of 90s

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- ❖ PC manufactures no longer labeled their units “Multimedia Computer”, because nearly every computer comes with multimedia capability
  - ❖ but R&D in multimedia continues
- ❖ R&D in Speech, Audio, Image and Video was fuelled mostly by telephony, broadcast and entertainment applications
  - ❖ Speech processing, compression, and recognition
  - ❖ Audio processing and compression
  - ❖ Image processing and compression
  - ❖ Video processing and compression



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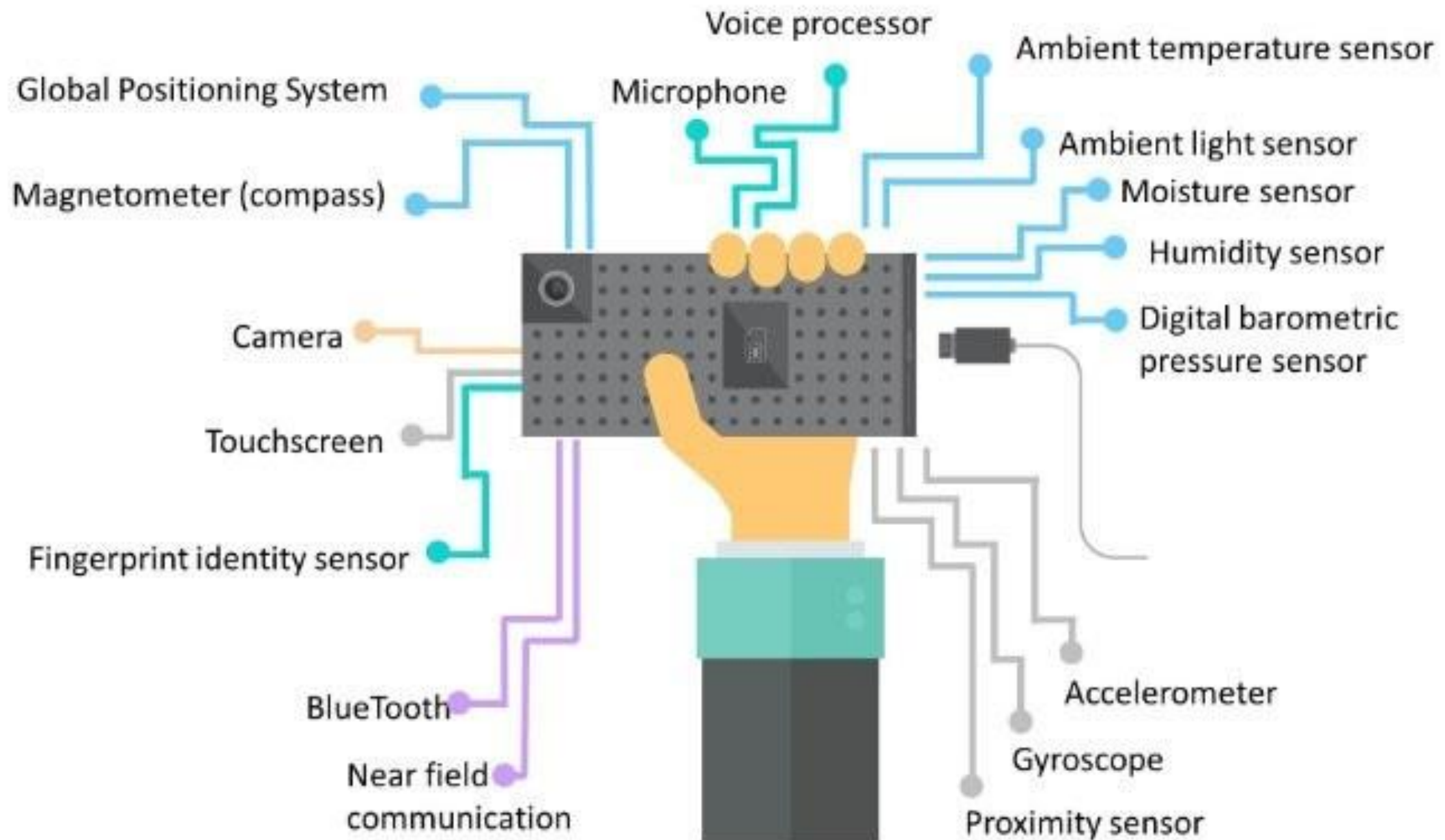
# The role of Computer in Multimedia in 21 century

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- ❖ Progress in processing, storage, networking, and software technology has changed computing beyond anyone's expectations
  - ❖ Even our mobile devices are several orders of magnitude more powerful and sophisticated than computers in late 80s
  - ❖ PCs and laptops are now being replaced by personal devices like tablets and mobile phones
- ❖ Mobile devices can use audio and visual mechanisms equally effectively as traditional alphanumeric computing
  - ❖ They are equipped with more diverse sensing mechanisms than humans have
    - ❖ e.g. GPS receivers to sense geo-location
  - ❖ These are truly multimedia computing and communication devices



# Different Sensors on Mobile Devices



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# Applications drove Computer Technologies Development

| Applications                                                | Technologies Developed                                                              | Device(s) Name            |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------|
| Numerical Computations                                      | Arithmetic Processing Unit (ALU)                                                    | Computer                  |
| Business Data Processing Applications                       | Alphanumeric Processing, Database, Centralised Server Technologies                  | Workstation, Server       |
| Word Processing, Spreadsheets, Mail, WWW                    | Internet Connectivity, More power CPU                                               | Personal Computer, Laptop |
| Social Network, Photography, Instant Chat and Media Sharing | 3G, 4G communications; Audio and Video Media Processing, Geo-Location Technologies. | Mobile Phones, Tablets    |
| Future Applications ???                                     | Future Technologies ???                                                             | ???                       |



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# Properties of Emerging Computing and Communication Applications

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- ❖ Spatiotemporal and live data streams are norm rather than the exception
- ❖ A holistic picture of an event or situation is more important than silos of isolated data
- ❖ Users want insights and information that are independent of the medium and data source
- ❖ Only relevant information counts
- ❖ Exploration (browsing) is the predominant mode of interaction
  - ❖ Instead of relational database querying is the primary interaction in previous generation application



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# Properties of Emerging Computing and Communication Applications (Cont'd)

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- ❖ Emerging applications are pushing
  - ❖ *Computing to use primarily data from multiple sources*
  - ❖ *Computing focus more on information, experiences, and understanding than on the medium or data source*
- ❖ Two aspects to pay attention
  - ❖ *The evolving nature of information*
  - ❖ *Expected results from computing*



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# E-com Operations Related Information

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- ❖ Question 1: *How many orders are operated by Amazon between Hong Kong and Tai wan?*
- ❖ Question 2: *Which parts of the world are seeing rapid growth in Amazon sales record?*
- ❖ Question 3: *What is the hottest item now?*
- ❖ What is/are the differences of these questions in terms of
  - ❖ *The evolving nature of information*
  - ❖ *Expected results from computing*



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# Experiential Environments

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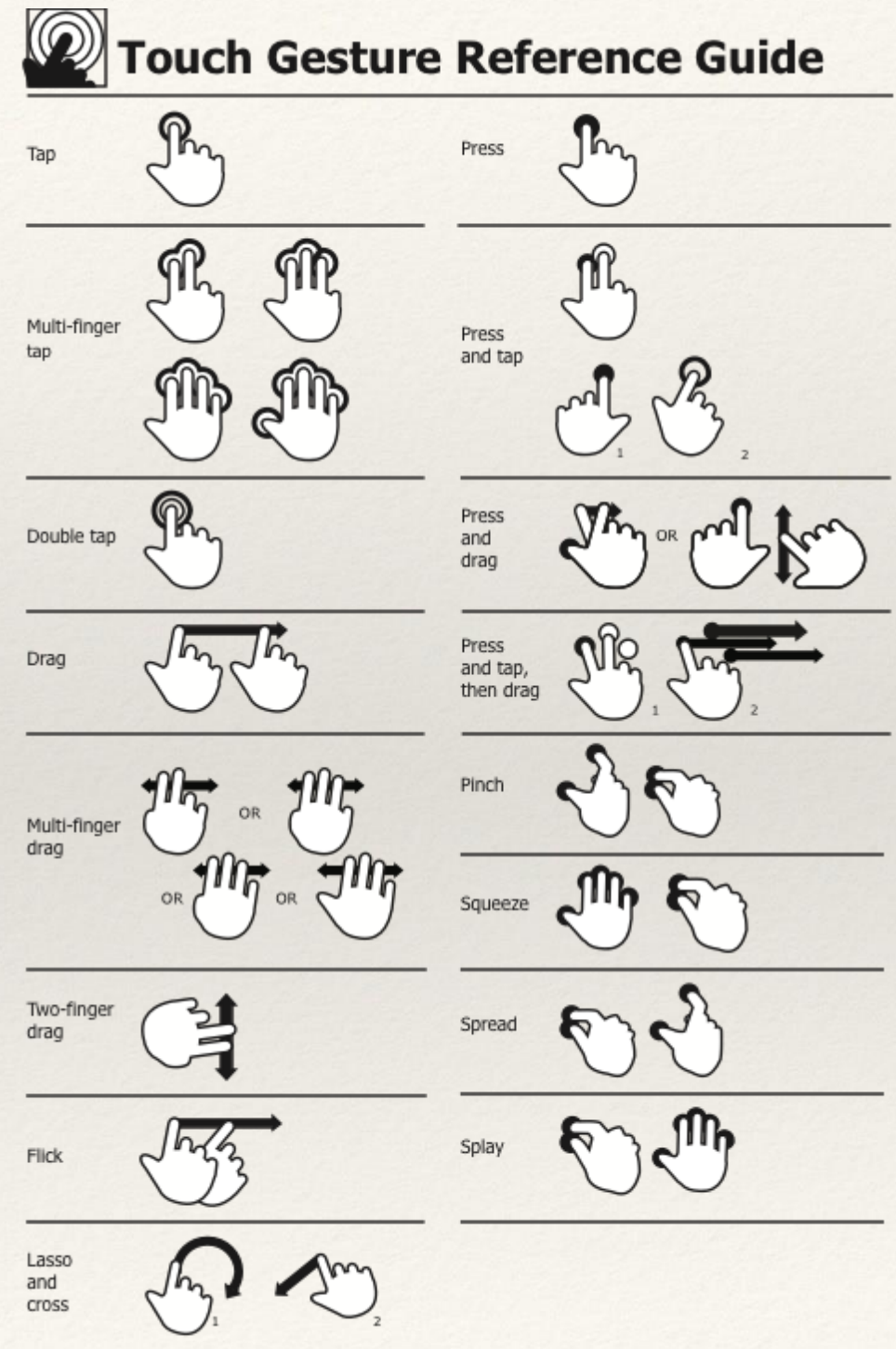
- ❖ Current information environments often work against the human-machine synergy
  - ❖ Humans are good at *conceptual and perceptual analysis*
  - ❖ Computers are exactly the opposite, they are efficient at *mathematical and logical analysis*
- ❖ Experiential environment is an environment where users directly use their senses to observe data and information of interest related to an event
  - ❖ Natural interaction with the data based on the users' particular set of interests in the *context* of that event



# Properties of Experiential Environments (1)

## ❖ *Direct and Intuitive*

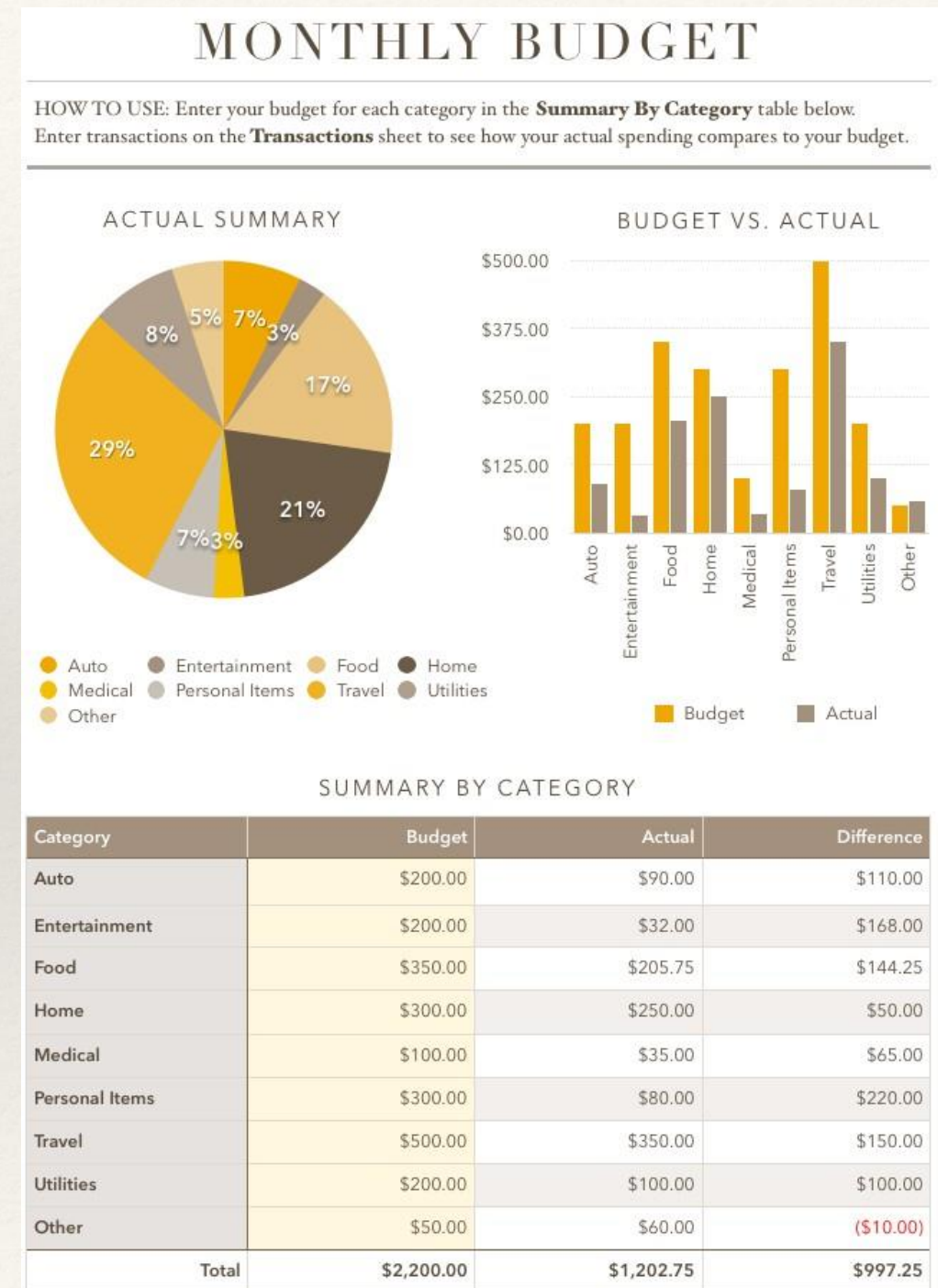
- ❖ Provides a holistic picture of an event without using unfamiliar metaphors and commands.
- ❖ Example(s)
  - ❖ Swiping a photo toward the left brings a new photo
  - ❖ Pushing on an icon representing an application launches an application
  - ❖ Game systems like Microsoft Kinect
- ❖ In experiential environments, users easily and rapidly interpret the data presented and then interact with the dataset to get a modified dataset



# Properties of Experiential Environments (2)

## ❖ *Provide the same query and presentation spaces*

- ❖ Query and presentation spaces are the same (WYSIWYG)
- ❖ Example(s)
  - ❖ Spreadsheet
    - ❖ Query: Change of certain data that is displayed in the context of other data items
    - ❖ Presentation: Results in a new sheet showing new relationships



# Properties of Experiential Environments (3)

- ❖ *Consider both the User State and Context*
  - ❖ People operate best when they are in known contexts
  - ❖ Good experiential environment should promote perceptual analysis and exploration
  - ❖ Example(s)
    - ❖ Video Games effectively use audio, video and tactile media to create compelling interactive environments



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# Why Multimedia Computing?

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- ❖ We need a good experiential systems to improve human-machine synergy for current information environments
- ❖ Experiential Systems are technically implemented as Multimedia Systems
  - ❖ Integrate, Process and Output data from different sensory modalities

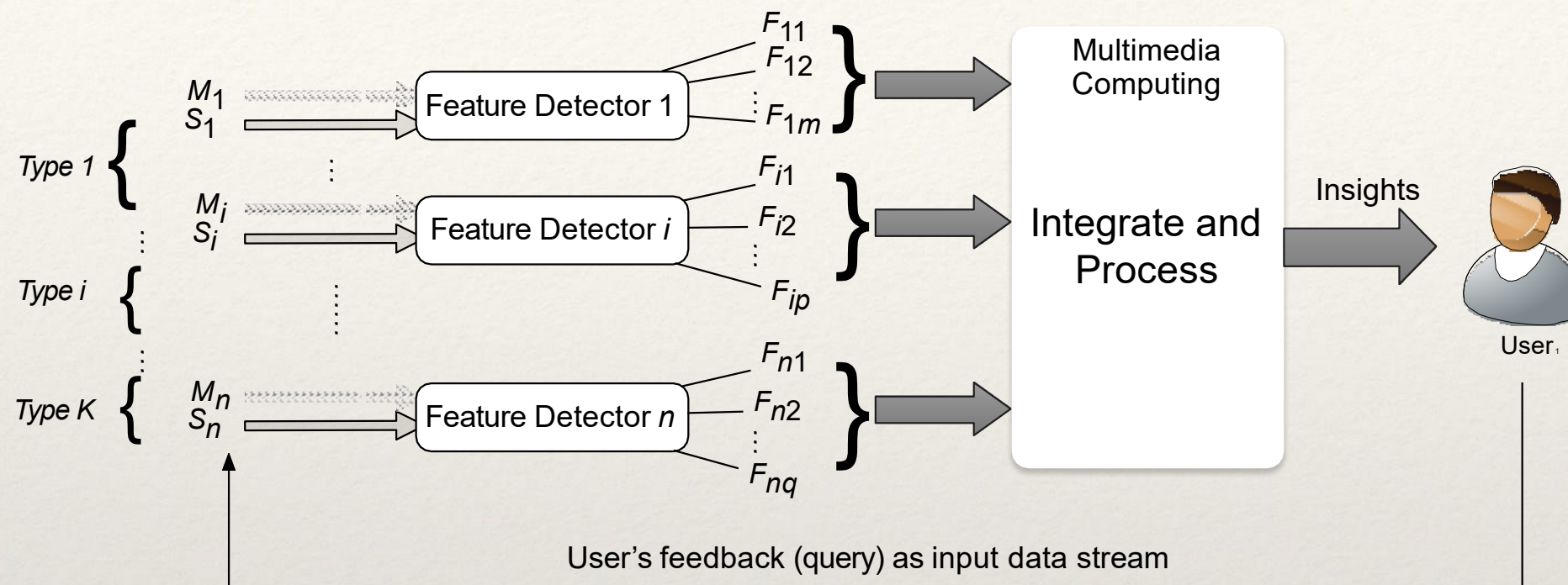


# Formal Definition of Multimedia Computing

- ❖ Consider a system equipped with multiple sensors working in a physical environment
  - ❖ Assume that  $S_1, \dots, S_n$  are synchronised data streams from sensors
  - ❖ These data streams have  $K$  types of data in the form of
    - ❖ Image sequence, Audio stream, Motion detector, etc.
  - ❖ Let  $M_1, \dots, M_n$  be the metadata for each stream
    - ❖ Annotations, Parameters, etc.
  - ❖ Let  $F_{ij}$  be the  $j$ -th feature stream from  $S_i$



# Formal Definition of Multimedia Computing (Cont'd)



- ❖ Multimedia differs from monomedia fields like computer vision or audio processing in the way that
  - ❖ Multimedia is closely related to how humans experience the world
  - ❖ Partial information from multiple media sources is correlated and combined to get complete information about the environment
  - ❖ Multimedia computing and communication is fundamentally about combining information from multiple sources in the context of the problem being solved

# Example of Multimedia Computing Application

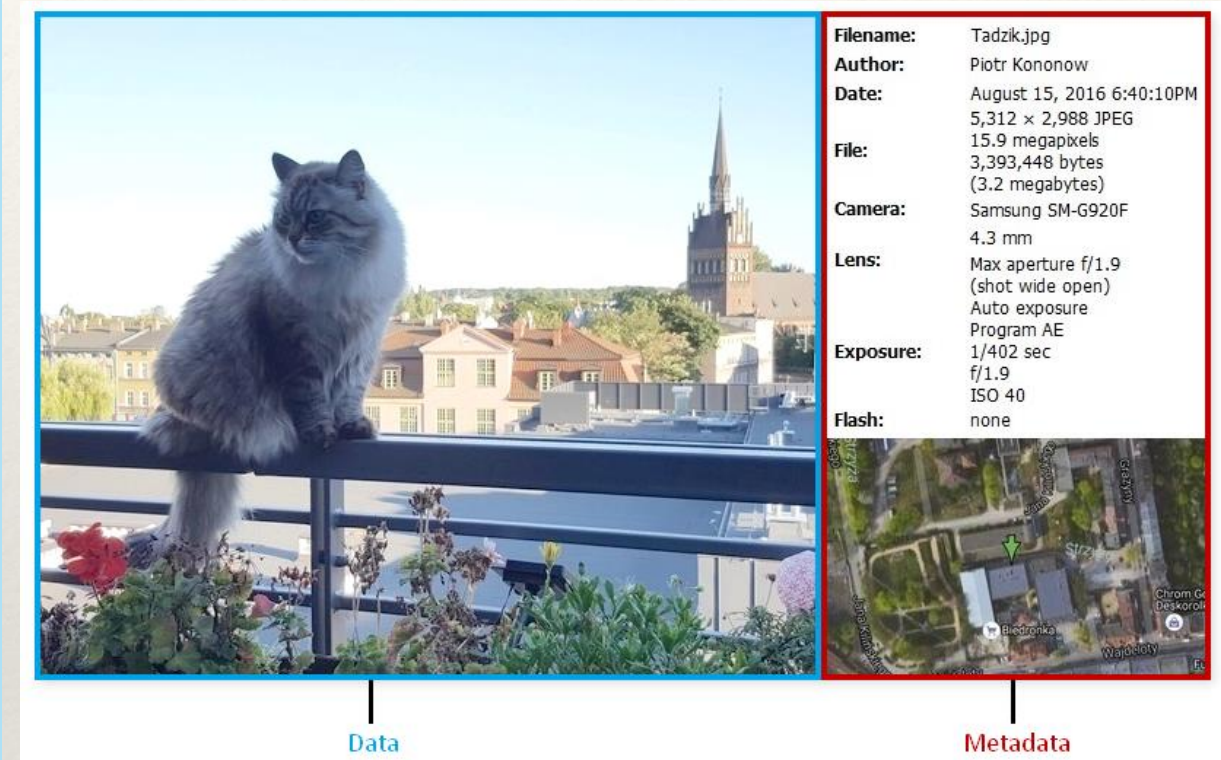
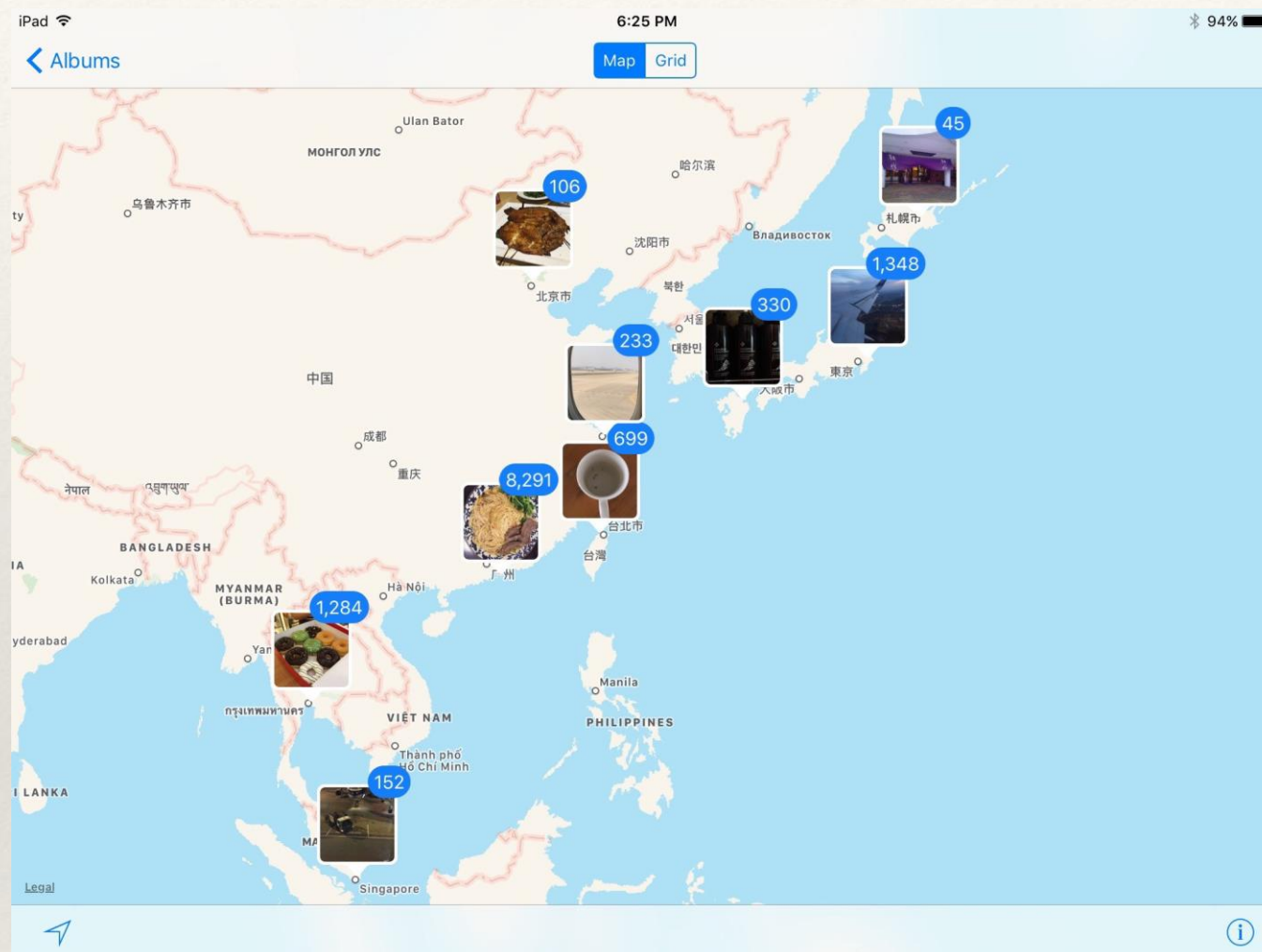


Image acquired from <https://dataedo.com/kb/data-glossary/what-is-metadata/>

Input: Metadata

Output: Insight (route of trip)